

Invited Workshop

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Attending to care in responsible and innovative design

Devices for teaching care ethics in Engineering Ethics Education

M. Nolan [a], S. Mahé [b], M.O. de Andrade [c], N. Barakat [d], S. Chance [e], S. Durán [f], A. Fujiki [g], A. Gwynne-Evans [h], S.J. Hitt [i],N. Al Kakoun [j], R. Kjelsberg [k], C.P. Leão [l], J. Lönngren [m], M. Makramalla [n], C. Shaw [o], S. Takehara [p], A.V. Tebeanu [q], M. Toboșaru [r] ,D.A. Martin [s],I. Villanueva [t]

[a] Atlantic Technological University Sligo, Ireland & University College Dublin SMARTlab, Inclusive Design Research Centre of Ireland, 0009-0003-8955-1481
[b] Ecole Normale Supérieure de Lyon, Lyon, France, 0009-0006-9544-8079
[c] Centre for Engineering Education, UCL, London, UK, 0000-0001-5682-0460
[d] University of Texas at Tyler, Tyler, TX, USA, 0000-0002-1622-1575
[e] Centre for Engineering Education, University College London, UK & School of Architecture, Building & Environment, Technological University Dublin, Ireland, 0000-0001-5598-7488
[f] Institut de Philosophie de Grenoble, Université Grenoble-Alpes, Grenoble, France, 0009-0000-9735-5059
[g] Kansai University, Suita, Osaka, Japan, 0000-0003-2302-9845
[h] Professional Communication Studies, Faculty of Engineering and the Built Environment, University of Cape Town, Cape Town, South Africa, 0000-0002-3737-7878
[i] NMITE, Hereford, UK, 0000-0002-0176-6214
[j] Engineering Learning Lab, American University of Beirut, Beirut, Lebanon, 0000-0001-9187-611X
[k] Department of Physics, NTNU – Norwegian University of Science and Technology, Trondheim, Norway, 0000-0001-8352-9038
[l] School of Engineering, University of Minho (UMinho), Portugal, 0000-0003-3725-5771
[m] Department of Science and Mathematics Education, Umeå University, Umeå, Sweden, 0000-0001-9667-2044
[n] NewGiza University, Giza, Egypt, 0000-0001-8151-3024
[o] Department of Mechanical Engineering, Faculty of Engineering and the Built Environment, University of Cape Town & Centre for Research in Engineering Education (CREE), Cape Town, South Africa, 0000-0002-9868-277X
[p] National Institute of Technology, Nara College, Japan, 0009-0001-5002-4436
[q] Department for Teacher Training and Social Sciences, National University of Science and Technology Politehnica Bucharest (UNSTPB), Romania 0009-0009-2876-1470
[r] Department for Teacher Training and Social Sciences, National University of Science and Technology Politehnica Bucharest (UNSTPB), Romania 0000-0002-1956-6747
[s]Centre for Engineering Education, University College London, London, UK, 0000-0002-9368-4100
[t] Department of Engineering Education, University of Florida, Gainesville, FL 32611

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ABSTRACT

Traditional engineering education has often prioritised technical problem-solving over relational, ethical, and holistic concerns. Recent work, including The Routledge International Handbook of Engineering Ethics Education, and seminal studies integrating qualitative data in experience-centered design highlight the need for more intentional integration of relational ethics and emotional awareness into engineering curricula.

In response, this workshop introduces the Design Thinking with Care framework, developed by the facilitator. Design thinking is increasingly used in engineering education and widely adopted in innovation practices. The framework maps Joan Tronto's phases of care - Attentiveness (caring about), Responsibility (taking care of), Competence (care giving), Responsiveness (care receiving), and Solidarity (caring with), onto the traditional stages of design thinking: Empathise, Define, Ideate, Prototype, and Test. The aim is to embed care-centered values more explicitly into design thinking stages.

This session focuses on the first stage of design: combining the Empathise stage of design thinking with Tronto's Caring About (attentiveness). Drawing on Jutta Treviranus' Inclusive Design principles, participants will complete an Inclusive Design Mapping exercise followed by a structured reflection on care, inclusion, and relational design priorities.

Through structured and interactive reflection, the workshop invites participants to notice lived experience, marginalisation, and exclusion, offering a starting point for cultivating methodically a relational and ethically aware approach to both design and pedagogy. It aims to spark small shifts in engineering education practices that can lead to broader ripple effects in the wider community, fostering care-based, inclusive thinking in engineering education and beyond. Participants will learn teaching devices to develop a transversal competence in engineering and ethics.

1 BACKGROUND AND RATIONALE

1.1 Limitations of Traditional Engineering Education

Traditional approaches to engineering often prioritise technical problem-solving, sometimes at the expense of relational, ethical, and inclusive considerations. Consequently, traditional approaches have relied on truncated views of experience-centred design. This emphasis on technical mastery has risked overlooking the lived, emotional, and relational realities of those affected by design decisions. Davis (Davis, 2017) and Roeser (Roeser, 2012) have argued for a more emotionally attuned approach to engineering, emphasising that emotions are not only relevant but essential to ethical reflection and responsible design. Their work supports

growing calls within engineering education to move beyond a purely rationalist paradigm and toward more human-centred, relationally grounded practices.

1.2 Emerging Perspectives on Relational Ethical and Emotional Awareness

Recent contributions to the field, including *The Routledge International Handbook of Engineering Ethics Education* (Chance et al., 2025), highlight the growing importance of integrating relational ethics, inclusion, and broader societal responsibilities into engineering education. A meta-synthesis by Lönngren et al. (2024) identifies empathy as a significant dimension in engineering education, conceptualised as part of emotional intelligence, professional responsibility, and relational care. However, the review also notes that engagement with emotions remains limited and often lacks theoretical grounding, reinforcing the need for more intentional, care-informed approaches to design and education.

1.3 The Design Thinking with Care Framework

The Design Thinking with Care framework responds to this gap by mapping Joan Tronto's five phases of care (Tronto, 1993, 2013), onto the traditional stages of design thinking (Auernhammer & Roth, 2021; Brown, 2008). The full Design Thinking with Care framework is presented in Nolan et al. (2025).

This workshop focuses on the first stage, Empathise, which aligns with Tronto's first phase of care: Caring about (attentiveness) (see Table 1). At this stage, empathy becomes an ethical practice, moving beyond observation to actively engaging with lived experience, noticing exclusion and marginalisation, and questioning how power and vulnerability shape the design context.

Informed by Treviranus' principles of Inclusive Design (Treviranus, 2023), this stage invites designers to intentionally address diversity and complexity from the outset. The workshop employs Inclusive Design Mapping (O'Neill, 2025) to explore a wide range of human factors — physical, sensory, cognitive, emotional, economic, geographic, cultural, linguistic, and temporal. By introducing a deeper focus on lived experience and relational attentiveness, participants are encouraged to reflect on their own roles, responsibilities, and assumptions within the design process, particularly in relation to those affected by engineering decisions.

Table 1: Empathise Stage in the Design Thinking with Care Framework

Design Thinking Stage	**Tronto's Phase of Care**	**Ethical Focus**	**Key Questions on Power & Vulnerability**
Empathise	Caring about (attentiveness)	Attending to lived experience; noticing exclusion and marginalisation	Whose experiences are being centred? Who is not seen or heard? Are users treated as full people or abstract personas?

2 WORKSHOP OBJECTIVES

2.1 Target audience

This workshop is open to educators and students interested in embedding relational, care-informed, and inclusive approaches into teaching, learning, and design. While most participants will be educators, it is equally relevant to students exploring how care ethics shapes practice and to researchers studying care in engineering

education. The aim is to create a ripple effect by sharing tools and fostering ethical awareness that extends beyond the classroom.

2.2 Expected learning outcomes

This workshop is intended to cultivate a shift in how educators approach design and teaching through a lens of care. While some participants may already engage with relational, care-informed practices, this session offers a shared space to reflect, map inclusive design approaches and explore the ethics of care, with a particular focus on "Caring About," the first phase of Joan Tronto's framework. Through attending to lived experience and noticing exclusion and marginalisation, the aim is to deepen our collective understanding of how care can shape both pedagogy and design.

By the end of this workshop, participants will:

1. Understand and apply methods for integrating ethics of care into their practice, such as attuning to their own and others' needs.
2. Reflect on personal and professional experiences to recognise care as a relational and ethical practice.
3. Identify dimensions of inclusion and exclusion and critically reflect on positionality, power, and participation through the Inclusive Design Mapping exercise.

3 WORKSHOP DESIGN

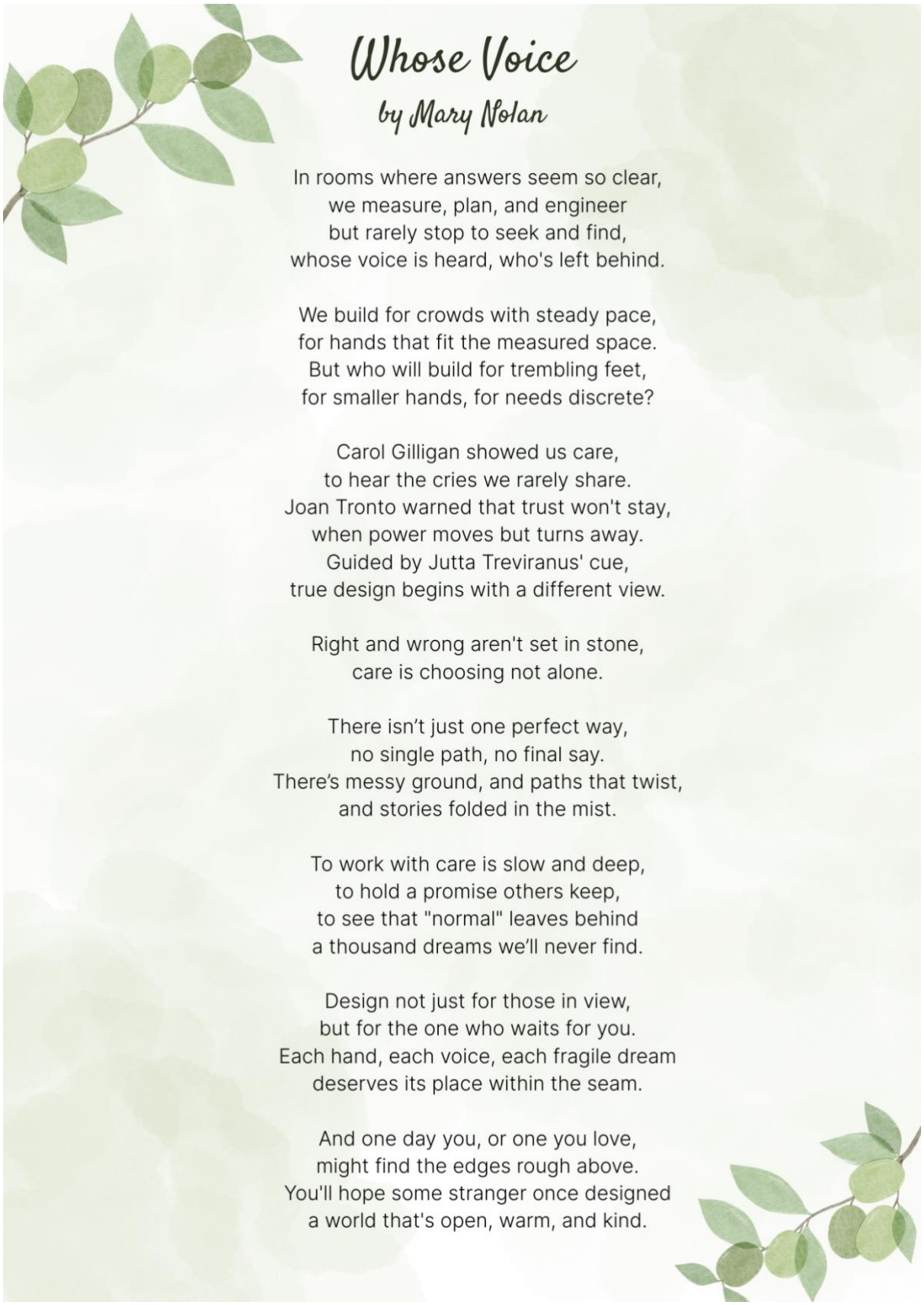
3.1 Time plan

The workshop is structured to guide participants through a series of reflective and participatory activities that build understanding of relational, care-informed design and teaching practices. Grounded in Tronto's ethics of care and the Design Thinking with Care framework, the session intentionally moves from personal reflection to broader engagement with inclusion, attentiveness, and ethical design. The workshop facilitator will introduce the guided reflection using a poem. The activities are designed to centre lived experience, surface positionality, and foster small, practical shifts in participants' own practice. Table 2 outlines the workshop structure, including timing, activity descriptions, and facilitation notes.

Table 2. Ethics of Care Workshop Time Plan

Run Time	Activity	Notes
0–10 mins	Welcome and Framing	Introduce aims, learning outcomes, and Tronto's phases of care; establish a reflective tone.
10-25 mins	Activity 1: Reflecting on Care	Guided reflection following the poem "Whose Voice?" (see Poem 1) to recall experiences of care.
25-40 mins	Activity 2: Attending to Lived Experience	Use of the Inclusive Design Mapping Tool (see appendix 1) to explore inclusion and exclusion across diverse human dimensions.
40-55mins	Activity 3: Seeds of Change	Individual and group reflection identifying one insight and one small shift to embed care-informed practice.
55-60 mins	Expansion Phase – Applying Care Ethics in Engineering	Summarise how care ethics can inform practice, invite continued engagement with care in engineering education, and conclude with encouragement.

Poem 1: Whose Voice by Mary Nolan 2025



Whose Voice

by Mary Nolan

In rooms where answers seem so clear,
we measure, plan, and engineer
but rarely stop to seek and find,
whose voice is heard, who's left behind.

We build for crowds with steady pace,
for hands that fit the measured space.
But who will build for trembling feet,
for smaller hands, for needs discrete?

Carol Gilligan showed us care,
to hear the cries we rarely share.
Joan Tronto warned that trust won't stay,
when power moves but turns away.
Guided by Jutta Treviranus' cue,
true design begins with a different view.

Right and wrong aren't set in stone,
care is choosing not alone.

There isn't just one perfect way,
no single path, no final say.
There's messy ground, and paths that twist,
and stories folded in the mist.

To work with care is slow and deep,
to hold a promise others keep,
to see that "normal" leaves behind
a thousand dreams we'll never find.

Design not just for those in view,
but for the one who waits for you.
Each hand, each voice, each fragile dream
deserves its place within the seam.

And one day you, or one you love,
might find the edges rough above.
You'll hope some stranger once designed
a world that's open, warm, and kind.

3.2 Interactivity

This workshop is designed to be highly interactive, encouraging participants to engage personally and collaboratively with the ideas presented. Participants will take part in a guided individual reflection, an inclusive design mapping exercise, and a group activity to reflect on key statements about care, inclusion, and relational design priorities. These activities are designed to move participants from personal reflection to relational awareness, and finally towards structured critical thinking about care-informed design practices. Opportunities for small group discussion and personal written reflection will support a participatory, reflective learning environment throughout the session.

4 WORKSHOP RESULTS

The workshop was designed to explore attuning to care, the first phase of the Design Thinking with Care framework, through a sequence of reflective and participatory activities. It was well attended, with five groups of three to four participants representing varied disciplinary and cultural backgrounds. This diversity enriched the discussion and demonstrated the value of collective reflection in engineering ethics education.

Reflecting on Care:
The session began with the poem “Whose Voice?” by Mary Nolan, which created an atmosphere of attentiveness and emotional openness. The guided reflection that followed prompted participants to recall a time they had given or received care. Responses ranged from deeply personal family experiences to reflections on care within teaching practice. Participants noted how storytelling deepened their understanding of care and helped them connect emotion to ethical awareness.

Attending to Lived Experience:
Participants then used the Inclusive Design Mapping Tool to explore inclusion and exclusion across physical, sensory, cognitive, social, and cultural factors. Although initially uncertain, all participants engaged fully, recognising how the tool revealed exclusions they had not previously considered. Discussion centred on the effort required to attune to others, with one participant remarking that, like tuning an old radio dial, it takes time and careful attention to find the right frequency for understanding others’ needs.

Seeds of Change:
In the final reflection, participants identified small but meaningful shifts they intended to make in their teaching or design practice. Feedback highlighted that the poem helped create a reflective atmosphere that made participation easier and more authentic.

Overall, the workshop met its aims, fostering relational and ethical awareness while showing that attuning to care can be taught as an embodied and reflective practice within engineering education.

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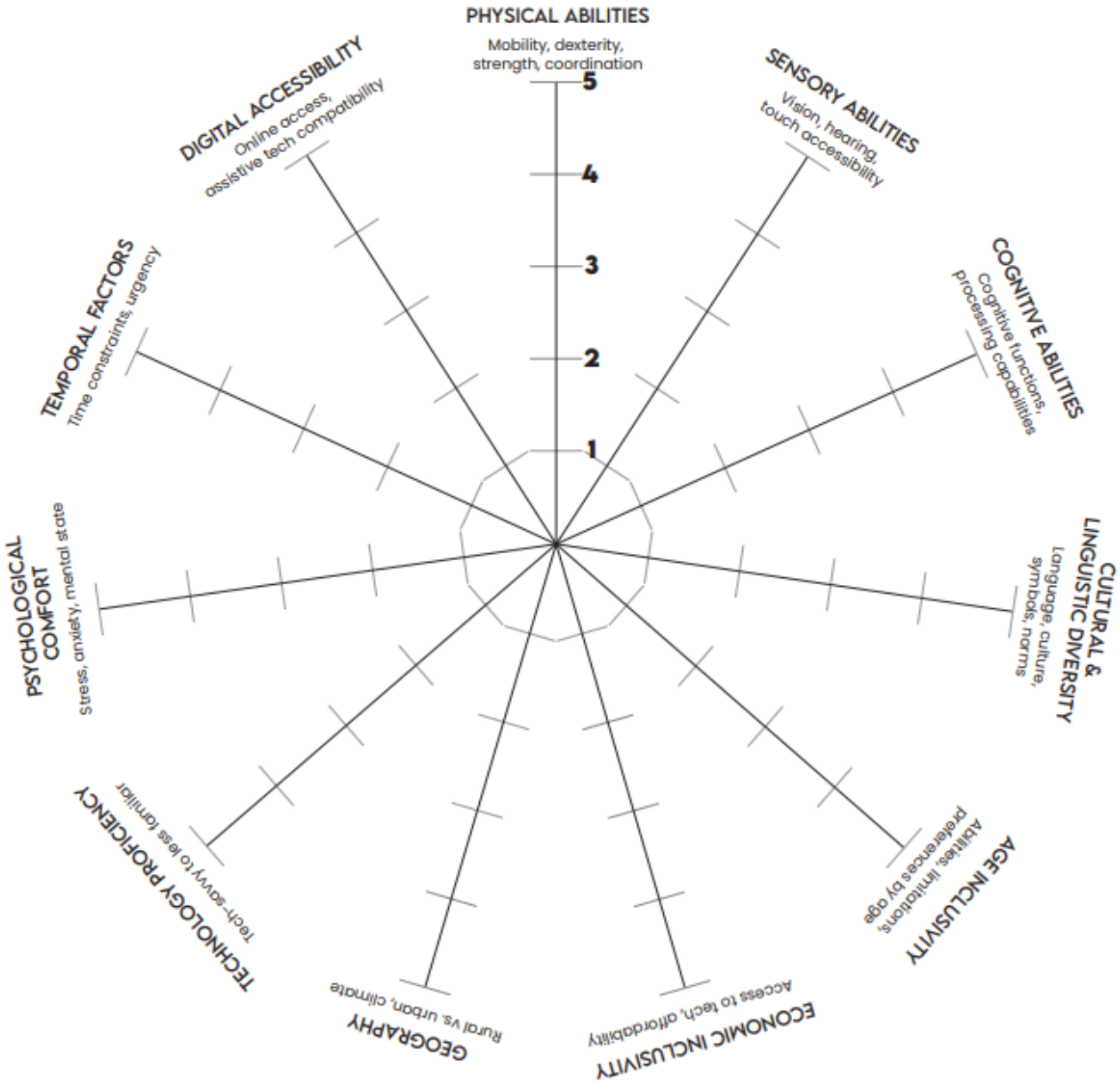
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Appendix 1

INCLUSIVE DESIGN MAPPING



LEGEND

USABILITY SCALE

5 **POOR USABILITY** Major usability issues.
4 **BELOW AVERAGE USABILITY** Significant usability issues.
3 **MODERATE USABILITY** Balanced mix of strengths and weaknesses.
2 **EXCELLENT USABILITY** Minimal usability issues, workable
1 **EXCEPTIONAL USABILITY** Flawless usability no issues

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